

潘佳佳 20220801 HW2

思路: 假设函数 $\varphi = \int \frac{dy}{dx} dx$

Q2 $\varphi(t)$ 在 $(-\infty, \infty)$ 连续, $\exists \varphi(0)$,

关系式 $\Rightarrow \varphi'(t) = \varphi(t)\varphi'(0)$,

$\exists \varphi(t+s) = \varphi(t)\varphi(s)$, 求函数 $\varphi(t)$

$\varphi' = k\varphi$

令 $t=s=0$, s.t. $\varphi(0) = \varphi(0)\varphi(0)$, if $\varphi(0) \neq 0$, then $\varphi(0) = 1$

$$\therefore \varphi'(t) = \lim_{s \rightarrow 0} \frac{\varphi(t+s) - \varphi(t)}{s} = \lim_{s \rightarrow 0} \frac{\varphi(t)\varphi(s) - \varphi(t)}{s} = \varphi(t) \lim_{s \rightarrow 0} \frac{\varphi(s) - 1}{s} \quad (\varphi'(0))$$

$$\therefore \varphi'(0) = \varphi(0) \lim_{s \rightarrow 0} \frac{\varphi(s) - 1}{s}$$

$$\therefore f'(0) = \lim_{x \rightarrow 0} \frac{f(x+s) - f(x)}{s} \Rightarrow \varphi(0) = \lim_{s \rightarrow 0} \frac{\varphi(0+s) - \varphi(0)}{s} = \lim_{s \rightarrow 0} \frac{\varphi(s) - 1}{s}$$

$$\therefore \varphi'(t) = \varphi(t)\varphi'(0) \quad \text{令 } \varphi'(0) = k$$

$$\frac{d\varphi}{dt} = k\varphi \Rightarrow \ln|\varphi| = kt + C, \text{ 代入 } \varphi(0) = 1$$

$$\ln|1| = 0 = k(0) + C, \therefore C = 0,$$

$$\therefore \varphi = e^{kt}$$

Q5 $\frac{dy}{dx} = P(x)y + Q(x)$ (-阶非齐次) 任两解之差 $\{y_1, y_2\}$

为 $\frac{dy}{dx} = P(x)y$ 之解

(1) $y' = P_y + Q$, (两解之差)

$$(y_1 - y_2)' = y_1' - y_2' = (P_y + Q) - (P_y + Q) = P(y_1 - y_2) \quad (Q=0)$$

通解构造

(2) $y \neq 0$, $\tilde{y}$ 为特解, 齐次方程通解 $c y$.

$$\text{任一解 } y - \tilde{y} = c y, \quad y = c y(\infty) + \tilde{y}(\infty)$$

叠加原理

(3) 齐次方程 $(y' = P_y, L(y) = y' - P_y)$ (算子为线性)

$$L(cy) = cL(y) = 0, \quad L(y_1 \pm y_2) = L(y_1) \pm L(y_2) = 0 \therefore L(cy) = cQ$$

$$\# \quad cQ \neq Q, \text{ unless } c=1,$$

0 常数可提取 (线性)
 $\circ L(y)$ 算子 $= 0$ (齐次)
 其中 $L(y) = y' - P_y$

齐次 $\Rightarrow$ 成倍放缩

$$\text{则 } \dots \Rightarrow L(cy) = (cy)'$$

$$= c(y' - P_y)$$

$$\therefore y' - P_y = Q$$

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Q1 恰当微分

$$(1) \underbrace{(x^2+y)}_M dx + \underbrace{(x-2y)}_N dy = 0$$

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1 \quad \text{恰当}$$

$$F(x,y) = \int (x^2+y) dx = \frac{1}{3}x^3 + xy + g(y)$$

$$F_y(x,y) = 0 + x + g'(y), \therefore F_y(x,y) = N = x - 2y$$

$$\therefore g'(y) = -2y, \quad g(y) = -y^2$$

$$\text{通解: } \frac{1}{3}x^3 + xy - y^2 = C$$

$$(2) \underbrace{(y-3x^2)}_M dx - \underbrace{(4y-x)}_N dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 1 \quad \text{恰当}$$

$$F(x,y) = \int (y-3x^2) dx = xy - x^3 + g(y)$$

$$F_y(x,y) = x + g'(y) = x - 4y, \therefore g'(y) = -4y, \quad g(y) = -2y^2$$

$$\text{通解: } xy - x^3 - 2y^2 = C$$

$$(3) \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{2xy}{(x-y)^2} \quad \text{恰当}$$

$$\textcircled{1} \therefore -d\left(\frac{xy}{x-y}\right) = \frac{y^2}{(x-y)^2} dx - \frac{x^2}{(x-y)^2} dy$$

$$\textcircled{2} \frac{1}{y} dy - \frac{1}{x} dx = d(\ln|y|) - d(\ln|x|) = d(\ln|\frac{y}{x}|)$$

$$\text{通解: } d\left(\ln\left|\frac{y}{x}\right| - \frac{xy}{x-y}\right) = 0$$

$$\ln\left|\frac{y}{x}\right| - \frac{xy}{x-y} = C$$

$$(4) \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2xy \quad \text{恰当}$$

$$F(x,y) = \int dx = 3x^2y^2 + x^2 + g(y)$$

$$F_y(x,y) = 6xy + g'(y), \therefore g'(y) = 2y$$

$$g(y) = y^2$$

$$\text{通解: } 3x^2y^2 + x^2 + y^2 = C$$

$$(5) \left(\frac{1}{y} \sin \frac{x}{y} - \frac{y}{x^2} \cos \frac{x}{y} + 1\right) dx + \left(\frac{1}{x} \cos \frac{x}{y} - \frac{x}{y^2} \sin \frac{x}{y} + \frac{1}{y}\right) dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 1$$

$$\textcircled{1} \frac{1}{y} \sin \frac{x}{y} dx - \frac{x}{y^2} \sin \frac{x}{y} dy = d(\cos \frac{x}{y})$$

$$\textcircled{2} \frac{1}{x} \cos \frac{x}{y} dy - \frac{y}{x^2} \cos \frac{x}{y} dx = d(\sin \frac{x}{y})$$

$$\textcircled{3} dx + \frac{1}{y} dy = d(x + \frac{1}{y})$$

$$\therefore \text{通解: } (x + \frac{1}{y}) + \cos \frac{x}{y} - \sin \frac{x}{y} = C$$

Q. (1) $2x(ye^{xy}-1)dx + e^{xy}dy = 0$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ ✓

$\int ye^{xy} - x^2 = C$

Q. 4) $ydx - xdy = (x^2+y^2)dx$

$\Rightarrow \frac{-(x^2+y^2)}{x^2+y^2} = dx$

$-d(\arctan \frac{y}{x}) = dx$

$\arctan \frac{y}{x} + x = C$

(7) $(y-x^2)dx - xdy = 0$

$\frac{dy}{dx} - \frac{1}{x}y = -x$

$\mu = e^{\int -\frac{1}{x}dx} = \frac{1}{x}$

$\mu(x) = \mu(x)$

$\frac{d}{dx}(\frac{y}{x}) = -1, \therefore \frac{y}{x} = -x + C$

$y = Cx - x^2$

Q. $(\cos x - y \sin x)dx + (y \sin x + x \cos y)dy = 0$

$\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} = 0 \Rightarrow \mu = e^{\int \cos y dy} = e^y$

$\frac{dy}{dx} + P(y)x = Q(x)$

$e^y(\quad) = e^y C$

$e^y(y \sin x + x \cos x - \sin x) = C$

Q. 5. (2) $M(x,y)dx + N(x,y)dy = 0$

$\mu M + yN \neq 0, \mu = \square$

(2)

$x \frac{\partial M}{\partial x} + y \frac{\partial N}{\partial y} = nM$

$x \frac{\partial N}{\partial x} + y \frac{\partial N}{\partial y} = nN$

μ is a function, $\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}$

L, R are not equal

$\frac{\partial}{\partial x}(\frac{N}{M+yN}) \Rightarrow x \frac{\partial M}{\partial x} = nM - y \frac{\partial M}{\partial y}$

$\therefore \mu = \frac{1}{xM+yN}$ # Dot

Q. 7. (2) $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = N f(x) - M g(y)$

$\mu = e^{\int \quad}$

$\frac{\partial M}{\partial x} = \mu \cdot f(x)$

$\frac{\partial N}{\partial y} = \mu \cdot g(y)$

Integrate: $\frac{\partial(\mu M)}{\partial y} - \frac{\partial(\mu N)}{\partial x}$

$\mu \frac{\partial M}{\partial y} + M \frac{\partial \mu}{\partial y} = \mu \frac{\partial M}{\partial x} + N \frac{\partial \mu}{\partial x}$

$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = N f(x) - M g(y)$

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$$(5) \quad u = xy, \quad x = uy, \quad dx = u dy + y du$$

$$\text{例: } (u y^2 e^u + y^2)(u dy + y du) - u y^2 e^u dy = 0$$

$$u dy + y(u e^u + 1) du = 0$$

$$(22) \quad \frac{dy}{y} + \frac{u e^u + 1}{u} du = 0$$

$$\Rightarrow \ln|yu| + e^u = C$$

$$\therefore \ln|x| + e^{\frac{x}{y}} = C$$

$$(6) \quad x \frac{dy}{dx} = 1 + \left(\frac{dy}{dx}\right)^2, \quad \exists (y')^2$$

$$p = \frac{dy}{dx}, \quad x p = 1 + p^2$$

$$x = p + \frac{1}{p}, \quad dy = p dx$$

$$dx = \left(1 - \frac{1}{p^2}\right) dp$$

$$dy = p \left(1 - \frac{1}{p^2}\right) dp$$

$$y = \frac{1}{2} p^2 - \ln|p| + C$$

~~$$\frac{dy}{dx} = e^{\frac{y}{x}} + \frac{y}{x}$$~~
~~$$u = \frac{y}{x}, \quad y' = u + x \frac{du}{dx}$$~~

~~$$x \frac{du}{dx} = e^u,$$~~
~~$$\therefore e^{-\frac{1}{x}} + \ln|x| = C$$~~

(划掉)

15题 (15)

$$(2) \quad y^2 \left[1 - \left(\frac{dy}{dx}\right)^2\right] = 1$$

$$\left(\frac{dy}{dx}\right)^2 = 1 - \frac{1}{y^2} = \frac{y^2 - 1}{y^2}$$

$$\frac{dy}{dx} = \pm \frac{\sqrt{y^2 - 1}}{y}$$

$$\int \frac{y}{\sqrt{y^2 - 1}} dy = \pm dx$$

$$y^2 - (x + C)^2 = 1 \quad (\text{双曲线})$$

$$(22) \quad \frac{dy}{dx} + e^{\frac{y}{x}} = 0$$

$$\text{例: } F(x, y) = 0, \quad p = \frac{dy}{dx}$$

$$x = p + e^p$$

$$dx = (1 + e^p) dp$$

$$dy = p dx$$

$$y = \frac{1}{2} p^2 + (p - 1)e^p + C$$

$$(80) \quad \frac{dy}{dx} = \frac{4x^2 - 2xy^3 + 2y}{3xy^2 - y^5 + xy^2}$$

$$\frac{M}{N} dx - \frac{N}{M} dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -6xy^2 \quad (1)$$

$$x^4 - xy^3 + x^2y^2 - y^3 = C$$

Q3 $R = kv$, $k = \text{const.}$

$$m \frac{dv}{dt} = -kv$$

$$\frac{dv}{dt} = -rv, \quad r = \frac{k}{m}$$

$$\int \frac{dv}{v} = \int -r dt$$

$$\ln v = -rt + C$$

$$v(t) = v_0 e^{-rt}$$

$$\therefore v(0) = 5, \quad \therefore v(t) = 5e^{-rt}$$

$$e^{-20r} = \frac{2}{5} = 0.4$$

$$t = 120, \quad v(120) = 5e^{-120r},$$

$$v(120) \approx 0.23 \text{ m/s}$$

$$\therefore \frac{1}{3} 2\pi \text{ ms}^{-1}, \quad v(120) = 0.23 \text{ m/s}$$